10th Grade Midterm Analysis of Decisions: C4–C5
Solutions

Contents

Evaluated criteria

C4 Applies the Maximin and Maximax criteria to make optimistic or conservative decisions without probabilities, selecting the alternative with the highest possible payoff or the best worst-case, interpreting risks or limitations.

C5 Constructs opportunity loss tables and applies the Minimax regret criterion to select the alternative that minimises the maximum opport

All payoffs are in thousands of dollars.
Regret = best payoff in the state − payoff.

1 Introduction

This section presents the general procedure for analysing a payoff table under uncertainty. The alternatives are denoted by A_i, the states of nature by S_j, and the payoff obtained by choosing alternative A_i when state S_j occurs by

$$P(A_i, S_j).$$

The analysis uses three decision criteria:

1. **Maximax:** choose the alternative with the largest possible payoff.
2. **Maximin:** choose the alternative with the largest worst-case payoff.
3. **Minimax Regret:** choose the alternative whose maximum opportunity loss is the smallest.

Because the payoffs in this section are represented symbolically, the final choice depends on the numerical values assigned to the expressions $P(A_i, S_j)$.

Problems:

1.1 General 2×2 Problem

A decision maker must choose between two alternatives, A_1 and A_2. Two states of nature, S_1 and S_2, may occur. The payoff associated with each alternative–state combination is represented by $P(A_i, S_j)$, where i identifies the alternative and j identifies the state of nature.

General payoff table

State of nature	Alternative A_1	Alternative A_2
State S_1	$P(A_1, S_1)$	$P(A_2, S_1)$
State S_2	$P(A_1, S_2)$	$P(A_2, S_2)$

Use Maximax, Maximin, and Minimax Regret to determine the decision rules for this general payoff table.

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

Step 1: Find the largest payoff for each alternative.

For A_1, compare its payoffs under S_1 and S_2:

$$M_1 = \max\{P(A_1, S_1), P(A_1, S_2)\}.$$

For A_2, compare its payoffs under S_1 and S_2:

$$M_2 = \max\{P(A_2, S_1), P(A_2, S_2)\}.$$

Therefore,

$$M_1 = \max\{P(A_1, S_1), P(A_1, S_2)\},$$
$$M_2 = \max\{P(A_2, S_1), P(A_2, S_2)\}.$$

Step 2: Apply the Maximax criterion.

Compare the largest payoffs M_1 and M_2:

$$M^* = \max\{M_1, M_2\}.$$

Equivalently,

$$M^* = \max\left\{\max\{P(A_1, S_1), P(A_1, S_2)\}, \max\{P(A_2, S_1), P(A_2, S_2)\}\right\}.$$

The Maximax decision is

$$A_{\text{Maximax}} = \arg\max_{i\in\{1,2\}} \left\{ \max_{j\in\{1,2\}} P(A_i, S_j) \right\}.$$

Thus, choose A_1 if $M_1 > M_2$, choose A_2 if $M_2 > M_1$, and report a tie if $M_1 = M_2$.

Step 3: Find the smallest payoff for each alternative.

For A_1,

$$m_1 = \min\{P(A_1, S_1), P(A_1, S_2)\}.$$

For A_2,

$$m_2 = \min\{P(A_2, S_1), P(A_2, S_2)\}.$$

Therefore,

$$m_1 = \min\{P(A_1, S_1), P(A_1, S_2)\},$$
$$m_2 = \min\{P(A_2, S_1), P(A_2, S_2)\}.$$

Step 4: Apply the Maximin criterion.

Compare the smallest payoffs m_1 and m_2:

$$m^* = \max\{m_1, m_2\}.$$

Equivalently,

$$m^* = \max\left\{\min\{P(A_1, S_1), P(A_1, S_2)\}, \min\{P(A_2, S_1), P(A_2, S_2)\}\right\}.$$

The Maximin decision is

$$A_{\text{Maximin}} = \arg\max_{i\in\{1,2\}} \left\{ \min_{j\in\{1,2\}} P(A_i, S_j) \right\}.$$

Thus, choose A_1 if $m_1 > m_2$, choose A_2 if $m_2 > m_1$, and report a tie if $m_1 = m_2$.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Step 1: Find the best payoff in each state of nature.

Under S_1, compare the payoffs of A_1 and A_2:

$$b_1 = \max\{P(A_1, S_1), P(A_2, S_1)\}.$$

Under S_2, compare the payoffs of A_1 and A_2:

$$b_2 = \max\{P(A_1, S_2), P(A_2, S_2)\}.$$

Therefore,

$$b_1 = \max\{P(A_1, S_1), P(A_2, S_1)\},$$
$$b_2 = \max\{P(A_1, S_2), P(A_2, S_2)\}.$$

Step 2: Calculate the regret for every alternative–state combination.

The regret associated with alternative A_i under state S_j is

$$r_{ij} = b_j - P(A_i, S_j).$$

Consequently,

$$r_{11} = b_1 - P(A_1, S_1), \quad r_{12} = b_2 - P(A_1, S_2),$$
$$r_{21} = b_1 - P(A_2, S_1), \quad r_{22} = b_2 - P(A_2, S_2).$$

The general regret table is therefore

Alternative	State S_1	State S_2
A_1	$b_1 - P(A_1, S_1)$	$b_2 - P(A_1, S_2)$
A_2	$b_1 - P(A_2, S_1)$	$b_2 - P(A_2, S_2)$

Step 3: Find the maximum regret for each alternative.

For A_1,

$$R_1 = \max\{b_1 - P(A_1, S_1), b_2 - P(A_1, S_2)\}.$$

For A_2,

$$R_2 = \max\{b_1 - P(A_2, S_1), b_2 - P(A_2, S_2)\}.$$

Thus,

$$R_1 = \max\{r_{11}, r_{12}\},$$
$$R_2 = \max\{r_{21}, r_{22}\}.$$

Step 4: Apply the Minimax Regret criterion.

Compare the maximum regrets R_1 and R_2:

$$R^* = \min\{R_1, R_2\}.$$

The Minimax Regret decision is

$$A_{\text{Minimax Regret}} = \arg\min_{i \in \{1,2\}} \left\{ \max_{j \in \{1,2\}} [b_j - P(A_i, S_j)] \right\}.$$

Thus, choose A_1 if $R_1 < R_2$, choose A_2 if $R_2 < R_1$, and report a tie if $R_1 = R_2$.

General decisions summary

Criterion	General decision rule
Maximax	$\arg\max_{i\in\{1,2\}} \max_{j\in\{1,2\}} P(A_i, S_j)$
Maximin	$\arg\max_{i\in\{1,2\}} \min_{j\in\{1,2\}} P(A_i, S_j)$
Minimax Regret	$\arg\min_{i\in\{1,2\}} \max_{j\in\{1,2\}} [b_j - P(A_i, S_j)]$

1.2 General 4×3 Problem

A decision maker must choose among three alternatives, A_1, A_2, and A_3. Four states of nature, S_1, S_2, S_3, and S_4, may occur. The payoff associated with each alternative–state combination is represented by $P(A_i, S_j)$.

General payoff table

State of nature	Alternative A_1	Alternative A_2	Alternative A_3
State S_1	$P(A_1, S_1)$	$P(A_2, S_1)$	$P(A_3, S_1)$
State S_2	$P(A_1, S_2)$	$P(A_2, S_2)$	$P(A_3, S_2)$
State S_3	$P(A_1, S_3)$	$P(A_2, S_3)$	$P(A_3, S_3)$
State S_4	$P(A_1, S_4)$	$P(A_2, S_4)$	$P(A_3, S_4)$

Use Maximax, Maximin, and Minimax Regret to determine the decision rules for this general payoff table.

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

Step 1: Find the largest payoff for each alternative.

For the three alternatives,

$$
\begin{aligned}
M_1 &= \max\{P(A_1, S_1), P(A_1, S_2), P(A_1, S_3), P(A_1, S_4)\},\\
M_2 &= \max\{P(A_2, S_1), P(A_2, S_2), P(A_2, S_3), P(A_2, S_4)\},\\
M_3 &= \max\{P(A_3, S_1), P(A_3, S_2), P(A_3, S_3), P(A_3, S_4)\}.
\end{aligned}
$$

Step 2: Apply the Maximax criterion.

Compare the three largest payoffs:

$$M^* = \max\{M_1, M_2, M_3\}.$$

Equivalently,

$$\begin{aligned}M^* = \max\{ &\max\{P(A_1,S_1),P(A_1,S_2),P(A_1,S_3),P(A_1,S_4)\},\\ &\max\{P(A_2,S_1),P(A_2,S_2),P(A_2,S_3),P(A_2,S_4)\},\\ &\max\{P(A_3,S_1),P(A_3,S_2),P(A_3,S_3),P(A_3,S_4)\}\}.\end{aligned}$$

The Maximax decision is

$$A_{\text{Maximax}} = \arg\max_{i\in\{1,2,3\}} \left\{ \max_{j\in\{1,2,3,4\}} P(A_i,S_j) \right\}.$$

Choose the alternative whose value M_i is the largest. If two or more alternatives have the same largest value, the Maximax criterion produces a tie.

Step 3: Find the smallest payoff for each alternative.

For the three alternatives,

$$\begin{aligned}m_1 &= \min\{P(A_1,S_1),P(A_1,S_2),P(A_1,S_3),P(A_1,S_4)\},\\ m_2 &= \min\{P(A_2,S_1),P(A_2,S_2),P(A_2,S_3),P(A_2,S_4)\},\\ m_3 &= \min\{P(A_3,S_1),P(A_3,S_2),P(A_3,S_3),P(A_3,S_4)\}.\end{aligned}$$

Step 4: Apply the Maximin criterion.

Compare the three smallest payoffs:

$$m^* = \max\{m_1,m_2,m_3\}.$$

Equivalently,

$$\begin{aligned}m^* = \max\{ &\min\{P(A_1,S_1),P(A_1,S_2),P(A_1,S_3),P(A_1,S_4)\},\\ &\min\{P(A_2,S_1),P(A_2,S_2),P(A_2,S_3),P(A_2,S_4)\},\\ &\min\{P(A_3,S_1),P(A_3,S_2),P(A_3,S_3),P(A_3,S_4)\}\}.\end{aligned}$$

The Maximin decision is

$$A_{\text{Maximin}} = \arg\max_{i\in\{1,2,3\}} \left\{ \min_{j\in\{1,2,3,4\}} P(A_i,S_j) \right\}.$$

Choose the alternative whose value m_i is the largest. If two or more alternatives have the same largest value, the Maximin criterion produces a tie.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Step 1: Find the best payoff in each state of nature.

For the four states,

$$\begin{aligned}b_1 &= \max\{P(A_1,S_1),P(A_2,S_1),P(A_3,S_1)\},\\ b_2 &= \max\{P(A_1,S_2),P(A_2,S_2),P(A_3,S_2)\},\\ b_3 &= \max\{P(A_1,S_3),P(A_2,S_3),P(A_3,S_3)\},\\ b_4 &= \max\{P(A_1,S_4),P(A_2,S_4),P(A_3,S_4)\}.\end{aligned}$$

Step 2: Calculate the regret for every alternative–state combination.

For every alternative A_i and state S_j, the regret is

$$r_{ij} = b_j - P(A_i,S_j).$$

The general regret table is therefore

Alternative	State S_1	State S_2	State S_3	State S_4
A_1	$b_1 - P(A_1, S_1)$	$b_2 - P(A_1, S_2)$	$b_3 - P(A_1, S_3)$	$b_4 - P(A_1, S_4)$
A_2	$b_1 - P(A_2, S_1)$	$b_2 - P(A_2, S_2)$	$b_3 - P(A_2, S_3)$	$b_4 - P(A_2, S_4)$
A_3	$b_1 - P(A_3, S_1)$	$b_2 - P(A_3, S_2)$	$b_3 - P(A_3, S_3)$	$b_4 - P(A_3, S_4)$

Step 3: Find the maximum regret for each alternative.

For A_1,

$$R_1 = \max\{b_1 - P(A_1, S_1), b_2 - P(A_1, S_2), b_3 - P(A_1, S_3), b_4 - P(A_1, S_4)\}.$$

For A_2,

$$R_2 = \max\{b_1 - P(A_2, S_1), b_2 - P(A_2, S_2), b_3 - P(A_2, S_3), b_4 - P(A_2, S_4)\}.$$

For A_3,

$$R_3 = \max\{b_1 - P(A_3, S_1), b_2 - P(A_3, S_2), b_3 - P(A_3, S_3), b_4 - P(A_3, S_4)\}.$$

Equivalently,

$$\begin{aligned} R_1 &= \max\{r_{11}, r_{12}, r_{13}, r_{14}\}, \\ R_2 &= \max\{r_{21}, r_{22}, r_{23}, r_{24}\}, \\ R_3 &= \max\{r_{31}, r_{32}, r_{33}, r_{34}\}. \end{aligned}$$

Step 4: Apply the Minimax Regret criterion.

Compare the maximum regrets:

$$R^* = \min\{R_1, R_2, R_3\}.$$

The Minimax Regret decision is

$$A_{\text{Minimax Regret}} = \arg\min_{i\in\{1,2,3\}} \left\{ \max_{j\in\{1,2,3,4\}} [b_j - P(A_i, S_j)] \right\}.$$

Choose the alternative whose value R_i is the smallest. If two or more alternatives have the same smallest value, the Minimax Regret criterion produces a tie.

General decisions summary

Criterion	General decision rule
Maximax	$\arg\max_{i\in\{1,2,3\}} \max_{j\in\{1,2,3,4\}} P(A_i, S_j)$
Maximin	$\arg\max_{i\in\{1,2,3\}} \min_{j\in\{1,2,3,4\}} P(A_i, S_j)$
Minimax Regret	$\arg\min_{i\in\{1,2,3\}} \max_{j\in\{1,2,3,4\}} [b_j - P(A_i, S_j)]$

2 Problems 2×2

Problems:
2.1. Food Truck Location (ABB)
2.2. Promotional Campaign (ABA)
2.3. Equipment Lease (AAA)

2.1 Food Truck Location

A food truck operator must choose between Location A and Location B. Demand can be strong or weak. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a location.

Payoff table

State of nature	Location A	Location B
Strong demand	100	60
Weak demand	0	50

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Location A}) = \max\{100, 0\} = 100,$$
$$\max(\text{Location B}) = \max\{60, 50\} = 60.$$
$$\max\{\max(\text{Location A}), \max(\text{Location B})\} = \max\{100, 60\} = 100.$$

The Maximax choice is Location A because it has the highest possible payoff.

$$\min(\text{Location A}) = \min\{100, 0\} = 0,$$
$$\min(\text{Location B}) = \min\{60, 50\} = 50.$$
$$\max\{\min(\text{Location A}), \min(\text{Location B})\} = \max\{0, 50\} = 50.$$

The Maximin choice is Location B because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{Strong demand:}\ \max\{100, 60\} = 100,$$
$$\text{Weak demand:}\ \max\{0, 50\} = 50.$$

Regret table (best payoff − payoff):

Alternative	Strong demand	Weak demand	Maximum regret
Location A	$100 - 100 = 0$	$50 - 0 = 50$	50
Location B	$100 - 60 = 40$	$50 - 50 = 0$	40

The Minimax Regret choice is Location B because it has the smallest maximum regret, 40.

Decisions summary

Criterion	Final decision
Maximax	Location A
Maximin	Location B
Minimax Regret	Location B

2.2 Promotional Campaign

A business must choose between Campaign A and Campaign B. Customer response can be strong or weak. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a campaign.

Payoff table

State of nature	Campaign A	Campaign B
Strong response	100	80
Weak response	20	30

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Campaign A}) = \max\{100, 20\} = 100,$$
$$\max(\text{Campaign B}) = \max\{80, 30\} = 80.$$
$$\max\{\max(\text{Campaign A}), \max(\text{Campaign B})\} = \max\{100, 80\} = 100.$$

The Maximax choice is Campaign A because it has the highest possible payoff.

$$\min(\text{Campaign A}) = \min\{100, 20\} = 20,$$
$$\min(\text{Campaign B}) = \min\{80, 30\} = 30.$$

$$\max\{\min(\text{Campaign A}), \min(\text{Campaign B})\} = \max\{20, 30\} = 30.$$

The Maximin choice is Campaign B because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{Strong response: } \max\{100, 80\} = 100,$$
$$\text{Weak response: } \max\{20, 30\} = 30.$$

Regret table (best payoff − payoff):

Alternative	Strong response	Weak response	Maximum regret
Campaign A	$100 - 100 = 0$	$30 - 20 = 10$	10
Campaign B	$100 - 80 = 20$	$30 - 30 = 0$	20

The Minimax Regret choice is Campaign A because it has the smallest maximum regret, 10.

Decisions summary

Criterion	Final decision
Maximax	Campaign A
Maximin	Campaign B
Minimax Regret	Campaign A

2.3 Equipment Lease

A company must choose between Lease A and Lease B. Market conditions can be favourable or unfavourable. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a lease.

Payoff table

State of nature	Lease A	Lease B
Favourable market	70	60
Unfavourable market	40	30

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Lease A}) = \max\{70, 40\} = 70,$$
$$\max(\text{Lease B}) = \max\{60, 30\} = 60.$$
$$\max\{\max(\text{Lease A}), \max(\text{Lease B})\} = \max\{70, 60\} = 70.$$

The Maximax choice is Lease A because it has the highest possible payoff.

$$\min(\text{Lease A}) = \min\{70, 40\} = 40,$$
$$\min(\text{Lease B}) = \min\{60, 30\} = 30.$$
$$\max\{\min(\text{Lease A}), \min(\text{Lease B})\} = \max\{40, 30\} = 40.$$

The Maximin choice is Lease A because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{Favourable market: } \max\{70, 60\} = 70,$$
$$\text{Unfavourable market: } \max\{40, 30\} = 40.$$

Regret table (best payoff − payoff):

Alternative	Favourable	Unfavourable	Maximum regret
Lease A	$70 - 70 = 0$	$40 - 40 = 0$	0
Lease B	$70 - 60 = 10$	$40 - 30 = 10$	10

The Minimax Regret choice is Lease A because it has the smallest maximum regret, 0.

Decisions summary

Criterion	Final decision
Maximax	Lease A
Maximin	Lease A
Minimax Regret	Lease A

3 Problems 2×3

Problems:

3.1 Production Plan

A manufacturer must choose Production Plan A, Plan B, or Plan C. Demand can be strong or weak. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a plan.

Payoff table

State of nature	Plan A	Plan B	Plan C
Strong demand	100	70	80
Weak demand	0	60	20

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Plan A}) = \max\{100, 0\} = 100,$$
$$\max(\text{Plan B}) = \max\{70, 60\} = 70,$$
$$\max(\text{Plan C}) = \max\{80, 20\} = 80.$$
$$\max\{\max(\text{Plan A}), \max(\text{Plan B}), \max(\text{Plan C})\} = \max\{100, 70, 80\} = 100.$$

The Maximax choice is Plan A because it has the highest possible payoff.

$$\min(\text{Plan A}) = \min\{100, 0\} = 0,$$
$$\min(\text{Plan B}) = \min\{70, 60\} = 60,$$
$$\min(\text{Plan C}) = \min\{80, 20\} = 20.$$
$$\max\{\min(\text{Plan A}), \min(\text{Plan B}), \min(\text{Plan C})\} = \max\{0, 60, 20\} = 60.$$

The Maximin choice is Plan B because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{Strong demand:}\ \max\{100, 70, 80\} = 100,$$
$$\text{Weak demand:}\ \max\{0, 60, 20\} = 60.$$

Regret table (best payoff − payoff):

Alternative	Strong demand	Weak demand	Maximum regret
Plan A	$100 - 100 = 0$	$60 - 0 = 60$	60
Plan B	$100 - 70 = 30$	$60 - 60 = 0$	30
Plan C	$100 - 80 = 20$	$60 - 20 = 40$	40

The Minimax Regret choice is Plan B because it has the smallest maximum regret, 30.

Decisions summary

Criterion	Final decision
Maximax	Plan A
Maximin	Plan B
Minimax Regret	Plan B

3.2 Marketing Strategy

A retailer must select Strategy A, Strategy B, or Strategy C for a new marketing campaign. Customer response can be strong or weak. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a strategy.

Payoff table

State of nature	Strategy A	Strategy B	Strategy C
Strong response	100	80	70
Weak response	40	50	45

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Strategy A}) = \max\{100, 40\} = 100,$$
$$\max(\text{Strategy B}) = \max\{80, 50\} = 80,$$
$$\max(\text{Strategy C}) = \max\{70, 45\} = 70.$$
$$\max\{\max(\text{Strategy A}), \max(\text{Strategy B}), \max(\text{Strategy C})\} = \max\{100, 80, 70\} = 100.$$

The Maximax choice is Strategy A because it has the highest possible payoff.

$$\min(\text{Strategy A}) = \min\{100, 40\} = 40,$$
$$\min(\text{Strategy B}) = \min\{80, 50\} = 50,$$
$$\min(\text{Strategy C}) = \min\{70, 45\} = 45.$$
$$\max\{\min(\text{Strategy A}), \min(\text{Strategy B}), \min(\text{Strategy C})\} = \max\{40, 50, 45\} = 50.$$

The Maximin choice is Strategy B because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{Strong response: } \max\{100, 80, 70\} = 100,$$
$$\text{Weak response: } \max\{40, 50, 45\} = 50.$$

Regret table (best payoff − payoff):

Alternative	Strong response	Weak response	Maximum regret
Strategy A	$100 - 100 = 0$	$50 - 40 = 10$	10
Strategy B	$100 - 80 = 20$	$50 - 50 = 0$	20
Strategy C	$100 - 70 = 30$	$50 - 45 = 5$	30

The Minimax Regret choice is Strategy A because it has the smallest maximum regret, 10.

Decisions summary

Criterion	Final decision
Maximax	Strategy A
Maximin	Strategy B
Minimax Regret	Strategy A

3.3 Event Venue

An event organiser must select Venue A, Venue B, or Venue C. Attendance can be high or low. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a venue.

Payoff table

State of nature	Venue A	Venue B	Venue C
High attendance	100	70	80
Low attendance	0	60	50

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Venue A}) = \max\{100, 0\} = 100,$$
$$\max(\text{Venue B}) = \max\{70, 60\} = 70,$$
$$\max(\text{Venue C}) = \max\{80, 50\} = 80.$$
$$\max\{\max(\text{Venue A}), \max(\text{Venue B}), \max(\text{Venue C})\} = \max\{100, 70, 80\} = 100.$$

The Maximax choice is Venue A because it has the highest possible payoff.

$$\min(\text{Venue A}) = \min\{100, 0\} = 0,$$
$$\min(\text{Venue B}) = \min\{70, 60\} = 60,$$
$$\min(\text{Venue C}) = \min\{80, 50\} = 50.$$
$$\max\{\min(\text{Venue A}), \min(\text{Venue B}), \min(\text{Venue C})\} = \max\{0, 60, 50\} = 60.$$

The Maximin choice is Venue B because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{High attendance: } \max\{100, 70, 80\} = 100,$$
$$\text{Low attendance: } \max\{0, 60, 50\} = 60.$$

Regret table (best payoff − payoff):

Alternative	High attendance	Low attendance	Maximum regret
Venue A	$100 - 100 = 0$	$60 - 0 = 60$	60
Venue B	$100 - 70 = 30$	$60 - 60 = 0$	30
Venue C	$100 - 80 = 20$	$60 - 50 = 10$	20

The Minimax Regret choice is Venue C because it has the smallest maximum regret, 20.

Decisions summary

Criterion	Final decision
Maximax	Venue A
Maximin	Venue B
Minimax Regret	Venue C

3.4 Distribution Plan

A company must select Distribution Plan A, Plan B, or Plan C. Market conditions can be favourable or unfavourable. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a distribution plan.

Payoff table

State of nature	Plan A	Plan B	Plan C
Favourable market	90	80	70
Unfavourable market	60	40	50

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Plan A}) = \max\{90, 60\} = 90,$$
$$\max(\text{Plan B}) = \max\{80, 40\} = 80,$$
$$\max(\text{Plan C}) = \max\{70, 50\} = 70.$$
$$\max\{\max(\text{Plan A}), \max(\text{Plan B}), \max(\text{Plan C})\} = \max\{90, 80, 70\} = 90.$$

The Maximax choice is Plan A because it has the highest possible payoff.

$$\min(\text{Plan A}) = \min\{90, 60\} = 60,$$
$$\min(\text{Plan B}) = \min\{80, 40\} = 40,$$
$$\min(\text{Plan C}) = \min\{70, 50\} = 50.$$
$$\max\{\min(\text{Plan A}), \min(\text{Plan B}), \min(\text{Plan C})\} = \max\{60, 40, 50\} = 60.$$

The Maximin choice is Plan A because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{Favourable market: } \max\{90, 80, 70\} = 90,$$
$$\text{Unfavourable market: } \max\{60, 40, 50\} = 60.$$

Regret table (best payoff − payoff):

Alternative	Favourable	Unfavourable	Maximum regret
Plan A	$90 - 90 = 0$	$60 - 60 = 0$	0
Plan B	$90 - 80 = 10$	$60 - 40 = 20$	20
Plan C	$90 - 70 = 20$	$60 - 50 = 10$	20

The Minimax Regret choice is Plan A because it has the smallest maximum regret, 0.

Decisions summary

Criterion	Final decision
Maximax	Plan A
Maximin	Plan A
Minimax Regret	Plan A

4 Problems 3×2

Problems:
4.1. Routing System (AAB)
4.2. Service Model (ABA)
4.3. Inventory Policy (ABB)
4.4. Logistics Contract (AAA)

4.1 Routing System

A shipping company must choose between System A and System B. Three states of nature represent different operating conditions. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a routing system.

Payoff table

State of nature	System A	System B
State S_1	100	70
State S_2	50	49
State S_3	50	90

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

$$\max(\text{System A}) = \max\{100, 50, 50\} = 100,$$

$$\max(\text{System B}) = \max\{70, 49, 90\} = 90.$$

$$\max\{\max(\text{System A}), \max(\text{System B})\} = \max\{100, 90\} = 100.$$

The Maximax choice is System A because it has the highest possible payoff.

$$\min(\text{System A}) = \min\{100, 50, 50\} = 50,$$
$$\min(\text{System B}) = \min\{70, 49, 90\} = 49.$$
$$\max\{\min(\text{System A}), \min(\text{System B})\} = \max\{50, 49\} = 50.$$

The Maximin choice is System A because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{State } S_1 : \max\{100, 70\} = 100,$$
$$\text{State } S_2 : \max\{50, 49\} = 50,$$
$$\text{State } S_3 : \max\{50, 90\} = 90.$$

Regret table (best payoff − payoff):

Alternative	S_1	S_2	S_3	Maximum regret
System A	$100 - 100 = 0$	$50 - 50 = 0$	$90 - 50 = 40$	40
System B	$100 - 70 = 30$	$50 - 49 = 1$	$90 - 90 = 0$	30

The Minimax Regret choice is System B because it has the smallest maximum regret, 30.

Decisions summary

Criterion	Final decision
Maximax	System A
Maximin	System A
Minimax Regret	System B

4.2 Service Model

A service company must choose between Service Model A and Service Model B. Demand can be high, moderate, or low. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a service model.

Payoff table

State of nature	Model A	Model B
High demand	100	80
Moderate demand	45	50
Low demand	40	50

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Model A}) = \max\{100, 45, 40\} = 100,$$
$$\max(\text{Model B}) = \max\{80, 50, 50\} = 80.$$
$$\max\{\max(\text{Model A}), \max(\text{Model B})\} = \max\{100, 80\} = 100.$$

The Maximax choice is Model A because it has the highest possible payoff.

$$\min(\text{Model A}) = \min\{100, 45, 40\} = 40,$$
$$\min(\text{Model B}) = \min\{80, 50, 50\} = 50.$$
$$\max\{\min(\text{Model A}), \min(\text{Model B})\} = \max\{40, 50\} = 50.$$

The Maximin choice is Model B because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\begin{aligned} \text{High demand:} \;& \max\{100, 80\} = 100, \\ \text{Moderate demand:} \;& \max\{45, 50\} = 50, \\ \text{Low demand:} \;& \max\{40, 50\} = 50. \end{aligned}$$

Regret table (best payoff − payoff):

Alternative	High	Moderate	Low	Maximum regret
Model A	$100 - 100 = 0$	$50 - 45 = 5$	$50 - 40 = 10$	10
Model B	$100 - 80 = 20$	$50 - 50 = 0$	$50 - 50 = 0$	20

The Minimax Regret choice is Model A because it has the smallest maximum regret, 10.

Decisions summary

Criterion	Final decision
Maximax	Model A
Maximin	Model B
Minimax Regret	Model A

4.3 Inventory Policy

A retailer must choose between Inventory Policy A and Inventory Policy B. Demand can be high, moderate, or low. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select an inventory policy.

Payoff table

State of nature	Policy A	Policy B
High demand	100	70
Moderate demand	20	60
Low demand	0	50

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Policy A}) = \max\{100, 20, 0\} = 100,$$
$$\max(\text{Policy B}) = \max\{70, 60, 50\} = 70.$$
$$\max\{\max(\text{Policy A}), \max(\text{Policy B})\} = \max\{100, 70\} = 100.$$

The Maximax choice is Policy A because it has the highest possible payoff.

$$\min(\text{Policy A}) = \min\{100, 20, 0\} = 0,$$
$$\min(\text{Policy B}) = \min\{70, 60, 50\} = 50.$$
$$\max\{\min(\text{Policy A}), \min(\text{Policy B})\} = \max\{0, 50\} = 50.$$

The Maximin choice is Policy B because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{High demand: } \max\{100, 70\} = 100,$$
$$\text{Moderate demand: } \max\{20, 60\} = 60,$$
$$\text{Low demand: } \max\{0, 50\} = 50.$$

Regret table (best payoff − payoff):

Alternative	High	Moderate	Low	Maximum regret
Policy A	$100 - 100 = 0$	$60 - 20 = 40$	$50 - 0 = 50$	50
Policy B	$100 - 70 = 30$	$60 - 60 = 0$	$50 - 50 = 0$	30

The Minimax Regret choice is Policy B because it has the smallest maximum regret, 30.

Decisions summary

Criterion	Final decision
Maximax	Policy A
Maximin	Policy B
Minimax Regret	Policy B

4.4 Logistics Contract

A company must choose between Logistics Contract A and Logistics Contract B. Market conditions can be favourable, stable, or unfavourable. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a logistics contract.

Payoff table

State of nature	Contract A	Contract B
Favourable market	90	80
Stable market	70	50
Unfavourable market	60	40

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Contract A}) = \max\{90, 70, 60\} = 90,$$
$$\max(\text{Contract B}) = \max\{80, 50, 40\} = 80.$$
$$\max\{\max(\text{Contract A}), \max(\text{Contract B})\} = \max\{90, 80\} = 90.$$

The Maximax choice is Contract A because it has the highest possible payoff.

$$\min(\text{Contract A}) = \min\{90, 70, 60\} = 60,$$
$$\min(\text{Contract B}) = \min\{80, 50, 40\} = 40.$$
$$\max\{\min(\text{Contract A}), \min(\text{Contract B})\} = \max\{60, 40\} = 60.$$

The Maximin choice is Contract A because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\begin{aligned}\text{Favourable market: } & \max\{90, 80\} = 90,\\ \text{Stable market: } & \max\{70, 50\} = 70,\\ \text{Unfavourable market: } & \max\{60, 40\} = 60.\end{aligned}$$

Regret table (best payoff − payoff):

Alternative	Favourable	Stable	Unfavourable	Maximum regret
Contract A	$90 - 90 = 0$	$70 - 70 = 0$	$60 - 60 = 0$	0
Contract B	$90 - 80 = 10$	$70 - 50 = 20$	$60 - 40 = 20$	20

The Minimax Regret choice is Contract A because it has the smallest maximum regret, 0.

Decisions summary

Criterion	Final decision
Maximax	Contract A
Maximin	Contract A
Minimax Regret	Contract A

5 Problems 3×3

Problems:

5.1 Technology Platform

A company must choose Technology Platform A, Platform B, or Platform C. Operating conditions can be favourable, stable, or disrupted. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a technology platform.

Payoff table

State of nature	Platform A	Platform B	Platform C
Favourable conditions	100	70	60
Stable conditions	50	49	40
Disrupted conditions	50	90	45

Questions/Tasks.

- **C4** Applies the Maximin and Maximax criteria
- **C5** Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Platform A}) = \max\{100, 50, 50\} = 100,$$
$$\max(\text{Platform B}) = \max\{70, 49, 90\} = 90,$$
$$\max(\text{Platform C}) = \max\{60, 40, 45\} = 60.$$
$$\max\{\max(\text{Platform A}), \max(\text{Platform B}), \max(\text{Platform C})\} = \max\{100, 90, 60\} = 100.$$

The Maximax choice is Platform A because it has the highest possible payoff.

$$\min(\text{Platform A}) = \min\{100, 50, 50\} = 50,$$
$$\min(\text{Platform B}) = \min\{70, 49, 90\} = 49,$$
$$\min(\text{Platform C}) = \min\{60, 40, 45\} = 40.$$
$$\max\{\min(\text{Platform A}), \min(\text{Platform B}), \min(\text{Platform C})\} = \max\{50, 49, 40\} = 50.$$

The Maximin choice is Platform A because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\begin{aligned} \text{Favourable conditions: } & \max\{100, 70, 60\} = 100, \\ \text{Stable conditions: } & \max\{50, 49, 40\} = 50, \\ \text{Disrupted conditions: } & \max\{50, 90, 45\} = 90. \end{aligned}$$

Regret table (best payoff − payoff):

Alternative	Favourable	Stable	Disrupted	Maximum regret
Platform A	$100 - 100 = 0$	$50 - 50 = 0$	$90 - 50 = 40$	40
Platform B	$100 - 70 = 30$	$50 - 49 = 1$	$90 - 90 = 0$	30
Platform C	$100 - 60 = 40$	$50 - 40 = 10$	$90 - 45 = 45$	45

The Minimax Regret choice is Platform B because it has the smallest maximum regret, 30.

Decisions summary

Criterion	Final decision
Maximax	Platform A
Maximin	Platform A
Minimax Regret	Platform B

5.2 Service Model

A service company must choose Service Model A, Model B, or Model C. Demand can be high, moderate, or low. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a service model.

Payoff table

State of nature	Model A	Model B	Model C
High demand	100	80	70
Moderate demand	45	50	35
Low demand	40	50	45

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Model A}) = \max\{100, 45, 40\} = 100,$$
$$\max(\text{Model B}) = \max\{80, 50, 50\} = 80,$$
$$\max(\text{Model C}) = \max\{70, 35, 45\} = 70.$$
$$\max\{\max(\text{Model A}), \max(\text{Model B}), \max(\text{Model C})\} = \max\{100, 80, 70\} = 100.$$

The Maximax choice is Model A because it has the highest possible payoff.

$$\min(\text{Model A}) = \min\{100, 45, 40\} = 40,$$
$$\min(\text{Model B}) = \min\{80, 50, 50\} = 50,$$
$$\min(\text{Model C}) = \min\{70, 35, 45\} = 35.$$
$$\max\{\min(\text{Model A}), \min(\text{Model B}), \min(\text{Model C})\} = \max\{40, 50, 35\} = 50.$$

The Maximin choice is Model B because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\begin{aligned} \text{High demand:}\ & \max\{100, 80, 70\} = 100, \\ \text{Moderate demand:}\ & \max\{45, 50, 35\} = 50, \\ \text{Low demand:}\ & \max\{40, 50, 45\} = 50. \end{aligned}$$

Regret table (best payoff − payoff):

Alternative	High	Moderate	Low	Maximum regret
Model A	$100 - 100 = 0$	$50 - 45 = 5$	$50 - 40 = 10$	10
Model B	$100 - 80 = 20$	$50 - 50 = 0$	$50 - 50 = 0$	20
Model C	$100 - 70 = 30$	$50 - 35 = 15$	$50 - 45 = 5$	30

The Minimax Regret choice is Model A because it has the smallest maximum regret, 10.

Decisions summary

Criterion	Final decision
Maximax	Model A
Maximin	Model B
Minimax Regret	Model A

5.3 Inventory Policy

A retailer must choose Inventory Policy A, Policy B, or Policy C. Demand can be high, moderate, or low. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select an inventory policy.

Payoff table

State of nature	Policy A	Policy B	Policy C
High demand	100	70	60
Moderate demand	20	60	40
Low demand	0	50	30

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Policy A}) = \max\{100, 20, 0\} = 100,$$
$$\max(\text{Policy B}) = \max\{70, 60, 50\} = 70,$$
$$\max(\text{Policy C}) = \max\{60, 40, 30\} = 60.$$
$$\max\{\max(\text{Policy A}), \max(\text{Policy B}), \max(\text{Policy C})\} = \max\{100, 70, 60\} = 100.$$

The Maximax choice is Policy A because it has the highest possible payoff.

$$\min(\text{Policy A}) = \min\{100, 20, 0\} = 0,$$
$$\min(\text{Policy B}) = \min\{70, 60, 50\} = 50,$$
$$\min(\text{Policy C}) = \min\{60, 40, 30\} = 30.$$
$$\max\{\min(\text{Policy A}), \min(\text{Policy B}), \min(\text{Policy C})\} = \max\{0, 50, 30\} = 50.$$

The Maximin choice is Policy B because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\begin{aligned}\text{High demand:}\ & \max\{100, 70, 60\} = 100,\\ \text{Moderate demand:}\ & \max\{20, 60, 40\} = 60,\\ \text{Low demand:}\ & \max\{0, 50, 30\} = 50.\end{aligned}$$

Regret table (best payoff − payoff):

Alternative	High	Moderate	Low	Maximum regret
Policy A	$100 - 100 = 0$	$60 - 20 = 40$	$50 - 0 = 50$	50
Policy B	$100 - 70 = 30$	$60 - 60 = 0$	$50 - 50 = 0$	30
Policy C	$100 - 60 = 40$	$60 - 40 = 20$	$50 - 30 = 20$	40

The Minimax Regret choice is Policy B because it has the smallest maximum regret, 30.

Decisions summary

Criterion	Final decision
Maximax	Policy A
Maximin	Policy B
Minimax Regret	Policy B

5.4 Event Venue

An event organiser must choose Venue A, Venue B, or Venue C. Attendance can be high, moderate, or low. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select an event venue.

Payoff table

State of nature	Venue A	Venue B	Venue C
High attendance	100	70	80
Moderate attendance	0	60	50
Low attendance	20	55	50

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

$$\begin{aligned}\max(\text{Venue A}) &= \max\{100, 0, 20\} = 100,\\ \max(\text{Venue B}) &= \max\{70, 60, 55\} = 70,\\ \max(\text{Venue C}) &= \max\{80, 50, 50\} = 80.\end{aligned}$$

$$\max\{\max(\text{Venue A}), \max(\text{Venue B}), \max(\text{Venue C})\} = \max\{100, 70, 80\} = 100.$$

The Maximax choice is Venue A because it has the highest possible payoff.

$$\begin{aligned}\min(\text{Venue A}) &= \min\{100, 0, 20\} = 0,\\ \min(\text{Venue B}) &= \min\{70, 60, 55\} = 55,\\ \min(\text{Venue C}) &= \min\{80, 50, 50\} = 50.\end{aligned}$$

$$\max\{\min(\text{Venue A}), \min(\text{Venue B}), \min(\text{Venue C})\} = \max\{0, 55, 50\} = 55.$$

The Maximin choice is Venue B because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\begin{aligned}\text{High attendance: }& \max\{100, 70, 80\} = 100,\\ \text{Moderate attendance: }& \max\{0, 60, 50\} = 60,\\ \text{Low attendance: }& \max\{20, 55, 50\} = 55.\end{aligned}$$

Regret table (best payoff − payoff):

Alternative	High	Moderate	Low	Maximum regret
Venue A	$100-100=0$	$60-0=60$	$55-20=35$	60
Venue B	$100-70=30$	$60-60=0$	$55-55=0$	30
Venue C	$100-80=20$	$60-50=10$	$55-50=5$	20

The Minimax Regret choice is Venue C because it has the smallest maximum regret, 20.

Decisions summary

Criterion	Final decision
Maximax	Venue A
Maximin	Venue B
Minimax Regret	Venue C

5.5 Distribution Plan

A company must choose Distribution Plan A, Plan B, or Plan C. Market conditions can be favourable, stable, or unfavourable. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a distribution plan.

Payoff table

State of nature	Plan A	Plan B	Plan C
Favourable market	90	80	70
Stable market	70	50	60
Unfavourable market	60	40	30

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Plan A}) = \max\{90, 70, 60\} = 90,$$

$$\max(\text{Plan B}) = \max\{80, 50, 40\} = 80,$$

$$\max(\text{Plan C}) = \max\{70, 60, 30\} = 70.$$

$$\max\{\max(\text{Plan A}), \max(\text{Plan B}), \max(\text{Plan C})\} = \max\{90, 80, 70\} = 90.$$

The Maximax choice is Plan A because it has the highest possible payoff.

$$\min(\text{Plan A}) = \min\{90, 70, 60\} = 60,$$
$$\min(\text{Plan B}) = \min\{80, 50, 40\} = 40,$$
$$\min(\text{Plan C}) = \min\{70, 60, 30\} = 30.$$
$$\max\{\min(\text{Plan A}), \min(\text{Plan B}), \min(\text{Plan C})\} = \max\{60, 40, 30\} = 60.$$

The Maximin choice is Plan A because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\begin{aligned} \text{Favourable market:} &\ \max\{90, 80, 70\} = 90, \\ \text{Stable market:} &\ \max\{70, 50, 60\} = 70, \\ \text{Unfavourable market:} &\ \max\{60, 40, 30\} = 60. \end{aligned}$$

Regret table (best payoff − payoff):

Alternative	Favourable	Stable	Unfavourable	Maximum regret
Plan A	$90 - 90 = 0$	$70 - 70 = 0$	$60 - 60 = 0$	0
Plan B	$90 - 80 = 10$	$70 - 50 = 20$	$60 - 40 = 20$	20
Plan C	$90 - 70 = 20$	$70 - 60 = 10$	$60 - 30 = 30$	30

The Minimax Regret choice is Plan A because it has the smallest maximum regret, 0.

Decisions summary

Criterion	Final decision
Maximax	Plan A
Maximin	Plan A
Minimax Regret	Plan A

6 Problems 4×2

Problems:
6.1. Production System (AAB)
6.2. Service Network (ABA)
6.3. Inventory Strategy (ABB)
6.4. Supply Contract (AAA)

6.1 Production System

A manufacturing company must choose between Production System A and Production System B. Four states of nature represent different operating conditions. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a production system.

Payoff table

State of nature	System A	System B
State S_1	100	70
State S_2	50	49
State S_3	50	90
State S_4	55	52

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

$$\max(\text{System A}) = \max\{100, 50, 50, 55\} = 100,$$
$$\max(\text{System B}) = \max\{70, 49, 90, 52\} = 90.$$
$$\max\{\max(\text{System A}), \max(\text{System B})\} = \max\{100, 90\} = 100.$$

The Maximax choice is System A because it has the highest possible payoff.

$$\min(\text{System A}) = \min\{100, 50, 50, 55\} = 50,$$
$$\min(\text{System B}) = \min\{70, 49, 90, 52\} = 49.$$
$$\max\{\min(\text{System A}), \min(\text{System B})\} = \max\{50, 49\} = 50.$$

The Maximin choice is System A because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{State } S_1 : \max\{100, 70\} = 100,$$
$$\text{State } S_2 : \max\{50, 49\} = 50,$$
$$\text{State } S_3 : \max\{50, 90\} = 90,$$
$$\text{State } S_4 : \max\{55, 52\} = 55.$$

Regret table (best payoff − payoff):

Alternative	S_1	S_2	S_3	S_4	Maximum regret
System A	$100 - 100 = 0$	$50 - 50 = 0$	$90 - 50 = 40$	$55 - 55 = 0$	40
System B	$100 - 70 = 30$	$50 - 49 = 1$	$90 - 90 = 0$	$55 - 52 = 3$	30

The Minimax Regret choice is System B because it has the smallest maximum regret, 30.

Decisions summary

Criterion	Final decision
Maximax	System A
Maximin	System A
Minimax Regret	System B

6.2 Service Network

A service company must choose between Service Network A and Service Network B. Four states of nature represent different demand conditions. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a service network.

Payoff table

State of nature	Network A	Network B
State S_1	100	80
State S_2	45	50
State S_3	40	50
State S_4	48	52

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Network A}) = \max\{100, 45, 40, 48\} = 100,$$
$$\max(\text{Network B}) = \max\{80, 50, 50, 52\} = 80.$$
$$\max\{\max(\text{Network A}), \max(\text{Network B})\} = \max\{100, 80\} = 100.$$

The Maximax choice is Network A because it has the highest possible payoff.

$$\min(\text{Network A}) = \min\{100, 45, 40, 48\} = 40,$$
$$\min(\text{Network B}) = \min\{80, 50, 50, 52\} = 50.$$
$$\max\{\min(\text{Network A}), \min(\text{Network B})\} = \max\{40, 50\} = 50.$$

The Maximin choice is Network B because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{State } S_1 : \max\{100, 80\} = 100,$$
$$\text{State } S_2 : \max\{45, 50\} = 50,$$
$$\text{State } S_3 : \max\{40, 50\} = 50,$$
$$\text{State } S_4 : \max\{48, 52\} = 52.$$

Regret table (best payoff − payoff):

Alternative	S_1	S_2	S_3	S_4	Maximum regret
Network A	$100 - 100 = 0$	$50 - 45 = 5$	$50 - 40 = 10$	$52 - 48 = 4$	10
Network B	$100 - 80 = 20$	$50 - 50 = 0$	$50 - 50 = 0$	$52 - 52 = 0$	20

The Minimax Regret choice is Network A because it has the smallest maximum regret, 10.

Decisions summary

Criterion	Final decision
Maximax	Network A
Maximin	Network B
Minimax Regret	Network A

6.3 Inventory Strategy

A retailer must choose between Inventory Strategy A and Inventory Strategy B. Four states of nature represent different demand conditions. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select an inventory strategy.

Payoff table

State of nature	Strategy A	Strategy B
State S_1	100	70
State S_2	20	60
State S_3	0	50
State S_4	25	55

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Strategy A}) = \max\{100, 20, 0, 25\} = 100,$$
$$\max(\text{Strategy B}) = \max\{70, 60, 50, 55\} = 70.$$
$$\max\{\max(\text{Strategy A}), \max(\text{Strategy B})\} = \max\{100, 70\} = 100.$$

The Maximax choice is Strategy A because it has the highest possible payoff.

$$\min(\text{Strategy A}) = \min\{100, 20, 0, 25\} = 0,$$
$$\min(\text{Strategy B}) = \min\{70, 60, 50, 55\} = 50.$$
$$\max\{\min(\text{Strategy A}), \min(\text{Strategy B})\} = \max\{0, 50\} = 50.$$

The Maximin choice is Strategy B because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{State } S_1 : \max\{100, 70\} = 100,$$
$$\text{State } S_2 : \max\{20, 60\} = 60,$$
$$\text{State } S_3 : \max\{0, 50\} = 50,$$
$$\text{State } S_4 : \max\{25, 55\} = 55.$$

Regret table (best payoff − payoff):

Alternative	S_1	S_2	S_3	S_4	Maximum regret
Strategy A	$100 - 100 = 0$	$60 - 20 = 40$	$50 - 0 = 50$	$55 - 25 = 30$	50
Strategy B	$100 - 70 = 30$	$60 - 60 = 0$	$50 - 50 = 0$	$55 - 55 = 0$	30

The Minimax Regret choice is Strategy B because it has the smallest maximum regret, 30.

Decisions summary

Criterion	Final decision
Maximax	Strategy A
Maximin	Strategy B
Minimax Regret	Strategy B

6.4 Supply Contract

A company must choose between Supply Contract A and Supply Contract B. Four states of nature represent different market and supply conditions. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a supply contract.

Payoff table

State of nature	Contract A	Contract B
State S_1	90	80
State S_2	70	50
State S_3	60	40
State S_4	65	55

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Contract A}) = \max\{90, 70, 60, 65\} = 90,$$
$$\max(\text{Contract B}) = \max\{80, 50, 40, 55\} = 80.$$
$$\max\{\max(\text{Contract A}), \max(\text{Contract B})\} = \max\{90, 80\} = 90.$$

The Maximax choice is Contract A because it has the highest possible payoff.

$$\min(\text{Contract A}) = \min\{90, 70, 60, 65\} = 60,$$
$$\min(\text{Contract B}) = \min\{80, 50, 40, 55\} = 40.$$
$$\max\{\min(\text{Contract A}), \min(\text{Contract B})\} = \max\{60, 40\} = 60.$$

The Maximin choice is Contract A because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{State } S_1: \max\{90, 80\} = 90,$$
$$\text{State } S_2: \max\{70, 50\} = 70,$$
$$\text{State } S_3: \max\{60, 40\} = 60,$$
$$\text{State } S_4: \max\{65, 55\} = 65.$$

Regret table (best payoff − payoff):

Alternative	S_1	S_2	S_3	S_4	Maximum regret
Contract A	$90 - 90 = 0$	$70 - 70 = 0$	$60 - 60 = 0$	$65 - 65 = 0$	0
Contract B	$90 - 80 = 10$	$70 - 50 = 20$	$60 - 40 = 20$	$65 - 55 = 10$	20

The Minimax Regret choice is Contract A because it has the smallest maximum regret, 0.

Decisions summary

Criterion	Final decision
Maximax	Contract A
Maximin	Contract A
Minimax Regret	Contract A

7 Problems 2×4

Problems:
7.1. Market Entry Strategy (ABA)
7.2. Production Plan (ABB)
7.3. Facility Location (ABC)
7.4. Supply Agreement (AAA)

7.1 Market Entry Strategy

A company must choose among Market Entry Strategy A, Strategy B, Strategy C, and Strategy D. Market conditions can be favourable or unfavourable. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a market entry strategy.

Payoff table

State of nature	Strategy A	Strategy B	Strategy C	Strategy D
Favourable market	30	60	50	0
Unfavourable market	100	40	10	30

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

$$\begin{aligned}
\max(\text{Strategy A}) &= \max\{30, 100\} = 100,\\
\max(\text{Strategy B}) &= \max\{60, 40\} = 60,\\
\max(\text{Strategy C}) &= \max\{50, 10\} = 50,\\
\max(\text{Strategy D}) &= \max\{0, 30\} = 30.
\end{aligned}$$

$$\begin{aligned}
&\max\{\max(\text{Strategy A}), \max(\text{Strategy B}), \max(\text{Strategy C}), \max(\text{Strategy D})\}\\
&\quad = \max\{100, 60, 50, 30\} = 100.
\end{aligned}$$

The Maximax choice is Strategy A because it has the highest possible payoff.

$$\begin{aligned}
\min(\text{Strategy A}) &= \min\{30, 100\} = 30,\\
\min(\text{Strategy B}) &= \min\{60, 40\} = 40,\\
\min(\text{Strategy C}) &= \min\{50, 10\} = 10,\\
\min(\text{Strategy D}) &= \min\{0, 30\} = 0.
\end{aligned}$$

$$\begin{aligned}
&\max\{\min(\text{Strategy A}), \min(\text{Strategy B}), \min(\text{Strategy C}), \min(\text{Strategy D})\}\\
&\quad = \max\{30, 40, 10, 0\} = 40.
\end{aligned}$$

The Maximin choice is Strategy B because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\begin{aligned}
\text{Favourable market:} &\quad \max\{30, 60, 50, 0\} = 60,\\
\text{Unfavourable market:} &\quad \max\{100, 40, 10, 30\} = 100.
\end{aligned}$$

Regret table (best payoff − payoff):

Alternative	Favourable	Unfavourable	Maximum regret
Strategy A	$60 - 30 = 30$	$100 - 100 = 0$	30
Strategy B	$60 - 60 = 0$	$100 - 40 = 60$	60
Strategy C	$60 - 50 = 10$	$100 - 10 = 90$	90
Strategy D	$60 - 0 = 60$	$100 - 30 = 70$	70

The Minimax Regret choice is Strategy A because it has the smallest maximum regret, 30.

Decisions summary

Criterion	Final decision
Maximax	Strategy A
Maximin	Strategy B
Minimax Regret	Strategy A

7.2 Production Plan

A manufacturing company must choose among Production Plan A, Plan B, Plan C, and Plan D. Demand can be high or low. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a production plan.

Payoff table

State of nature	Plan A	Plan B	Plan C	Plan D
High demand	100	90	40	30
Low demand	70	90	80	40

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

$$\begin{aligned}
\max(\text{Plan A}) &= \max\{100, 70\} = 100,\\
\max(\text{Plan B}) &= \max\{90, 90\} = 90,\\
\max(\text{Plan C}) &= \max\{40, 80\} = 80,\\
\max(\text{Plan D}) &= \max\{30, 40\} = 40.
\end{aligned}$$

$$\begin{aligned}
&\max\{\max(\text{Plan A}), \max(\text{Plan B}), \max(\text{Plan C}), \max(\text{Plan D})\}\\
&\quad = \max\{100, 90, 80, 40\} = 100.
\end{aligned}$$

The Maximax choice is Plan A because it has the highest possible payoff.

$$\begin{aligned}
\min(\text{Plan A}) &= \min\{100, 70\} = 70,\\
\min(\text{Plan B}) &= \min\{90, 90\} = 90,\\
\min(\text{Plan C}) &= \min\{40, 80\} = 40,\\
\min(\text{Plan D}) &= \min\{30, 40\} = 30.
\end{aligned}$$

$$\begin{aligned}&\max\{\min(\text{Plan A}), \min(\text{Plan B}), \min(\text{Plan C}), \min(\text{Plan D})\}\\ &= \max\{70, 90, 40, 30\} = 90.\end{aligned}$$

The Maximin choice is Plan B because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\begin{aligned}\text{High demand:}\ & \max\{100, 90, 40, 30\} = 100,\\ \text{Low demand:}\ & \max\{70, 90, 80, 40\} = 90.\end{aligned}$$

Regret table (best payoff − payoff):

Alternative	High demand	Low demand	Maximum regret
Plan A	$100 - 100 = 0$	$90 - 70 = 20$	20
Plan B	$100 - 90 = 10$	$90 - 90 = 0$	10
Plan C	$100 - 40 = 60$	$90 - 80 = 10$	60
Plan D	$100 - 30 = 70$	$90 - 40 = 50$	70

The Minimax Regret choice is Plan B because it has the smallest maximum regret, 10.

Decisions summary

Criterion	Final decision
Maximax	Plan A
Maximin	Plan B
Minimax Regret	Plan B

7.3 Facility Location

A company must choose among Facility Location A, Location B, Location C, and Location D. Regional demand can be high or low. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a facility location.

Payoff table

State of nature	Location A	Location B	Location C	Location D
High regional demand	10	60	50	20
Low regional demand	100	70	80	40

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Location A}) = \max\{10, 100\} = 100,$$
$$\max(\text{Location B}) = \max\{60, 70\} = 70,$$
$$\max(\text{Location C}) = \max\{50, 80\} = 80,$$
$$\max(\text{Location D}) = \max\{20, 40\} = 40.$$

$$\max\{\max(\text{Location A}), \max(\text{Location B}), \max(\text{Location C}), \max(\text{Location D})\}$$
$$= \max\{100, 70, 80, 40\} = 100.$$

The Maximax choice is Location A because it has the highest possible payoff.

$$\min(\text{Location A}) = \min\{10, 100\} = 10,$$
$$\min(\text{Location B}) = \min\{60, 70\} = 60,$$
$$\min(\text{Location C}) = \min\{50, 80\} = 50,$$
$$\min(\text{Location D}) = \min\{20, 40\} = 20.$$

$$\max\{\min(\text{Location A}), \min(\text{Location B}), \min(\text{Location C}), \min(\text{Location D})\}$$
$$= \max\{10, 60, 50, 20\} = 60.$$

The Maximin choice is Location B because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{High regional demand: } \max\{10, 60, 50, 20\} = 60,$$
$$\text{Low regional demand: } \max\{100, 70, 80, 40\} = 100.$$

Regret table (best payoff $-$ payoff):

Alternative	High demand	Low demand	Maximum regret
Location A	$60 - 10 = 50$	$100 - 100 = 0$	50
Location B	$60 - 60 = 0$	$100 - 70 = 30$	30
Location C	$60 - 50 = 10$	$100 - 80 = 20$	20
Location D	$60 - 20 = 40$	$100 - 40 = 60$	60

The Minimax Regret choice is Location C because it has the smallest maximum regret, 20.

Decisions summary

Criterion	Final decision
Maximax	Location A
Maximin	Location B
Minimax Regret	Location C

7.4 Supply Agreement

A company must choose among Supply Agreement A, Agreement B, Agreement C, and Agreement D. Supply conditions can be stable or disrupted. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a supply agreement.

Payoff table

State of nature	Agreement A	Agreement B	Agreement C	Agreement D
Stable supply	90	70	60	30
Disrupted supply	100	80	50	70

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

$$\begin{aligned}
\max(\text{Agreement A}) &= \max\{90, 100\} = 100,\\
\max(\text{Agreement B}) &= \max\{70, 80\} = 80,\\
\max(\text{Agreement C}) &= \max\{60, 50\} = 60,\\
\max(\text{Agreement D}) &= \max\{30, 70\} = 70.
\end{aligned}$$

$$\begin{aligned}
&\max\{\max(\text{Agreement A}), \max(\text{Agreement B}), \max(\text{Agreement C}), \max(\text{Agreement D})\}\\
&\quad = \max\{100, 80, 60, 70\} = 100.
\end{aligned}$$

The Maximax choice is Agreement A because it has the highest possible payoff.

$$\begin{aligned}
\min(\text{Agreement A}) &= \min\{90, 100\} = 90,\\
\min(\text{Agreement B}) &= \min\{70, 80\} = 70,\\
\min(\text{Agreement C}) &= \min\{60, 50\} = 50,\\
\min(\text{Agreement D}) &= \min\{30, 70\} = 30.
\end{aligned}$$

$$\max\{\min(\text{Agreement A}), \min(\text{Agreement B}), \min(\text{Agreement C}), \min(\text{Agreement D})\}$$
$$= \max\{90, 70, 50, 30\} = 90.$$

The Maximin choice is Agreement A because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{Stable supply:} \ \max\{90, 70, 60, 30\} = 90,$$
$$\text{Disrupted supply:} \ \max\{100, 80, 50, 70\} = 100.$$

Regret table (best payoff − payoff):

Alternative	Stable supply	Disrupted supply	Maximum regret
Agreement A	$90 - 90 = 0$	$100 - 100 = 0$	0
Agreement B	$90 - 70 = 20$	$100 - 80 = 20$	20
Agreement C	$90 - 60 = 30$	$100 - 50 = 50$	50
Agreement D	$90 - 30 = 60$	$100 - 70 = 30$	60

The Minimax Regret choice is Agreement A because it has the smallest maximum regret, 0.

Decisions summary

Criterion	Final decision
Maximax	Agreement A
Maximin	Agreement A
Minimax Regret	Agreement A

8 Problems 4×3

Problems:

8.1 Production System

A manufacturing company must choose among Production System A, System B, and System C. Four states of nature represent different operating conditions. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a production system.

Payoff table

State of nature	System A	System B	System C
State S_1	100	70	60
State S_2	50	49	30
State S_3	50	90	40
State S_4	55	52	35

Questions/Tasks.

- **C4** Applies the Maximin and Maximax criteria
- **C5** Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

$$\begin{aligned}\max(\text{System A}) &= \max\{100, 50, 50, 55\} = 100,\\ \max(\text{System B}) &= \max\{70, 49, 90, 52\} = 90,\\ \max(\text{System C}) &= \max\{60, 30, 40, 35\} = 60.\end{aligned}$$

$$\begin{aligned}&\max\{\max(\text{System A}), \max(\text{System B}), \max(\text{System C})\}\\ &= \max\{100, 90, 60\} = 100.\end{aligned}$$

The Maximax choice is System A because it has the highest possible payoff.

$$\begin{aligned}\min(\text{System A}) &= \min\{100, 50, 50, 55\} = 50,\\ \min(\text{System B}) &= \min\{70, 49, 90, 52\} = 49,\\ \min(\text{System C}) &= \min\{60, 30, 40, 35\} = 30.\end{aligned}$$

$$\begin{aligned}&\max\{\min(\text{System A}), \min(\text{System B}), \min(\text{System C})\}\\ &= \max\{50, 49, 30\} = 50.\end{aligned}$$

The Maximin choice is System A because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\begin{aligned}\text{State } S_1 &: \max\{100, 70, 60\} = 100,\\ \text{State } S_2 &: \max\{50, 49, 30\} = 50,\\ \text{State } S_3 &: \max\{50, 90, 40\} = 90,\\ \text{State } S_4 &: \max\{55, 52, 35\} = 55.\end{aligned}$$

Regret table (best payoff − payoff):

Alternative	S_1	S_2	S_3	S_4	Maximum regret
System A	$100 - 100 = 0$	$50 - 50 = 0$	$90 - 50 = 40$	$55 - 55 = 0$	40
System B	$100 - 70 = 30$	$50 - 49 = 1$	$90 - 90 = 0$	$55 - 52 = 3$	30
System C	$100 - 60 = 40$	$50 - 30 = 20$	$90 - 40 = 50$	$55 - 35 = 20$	50

The Minimax Regret choice is System B because it has the smallest maximum regret, 30.

Decisions summary

Criterion	Final decision
Maximax	System A
Maximin	System A
Minimax Regret	System B

8.2 Service Network

A service company must choose among Service Network A, Network B, and Network C. Four states of nature represent different demand conditions. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a service network.

Payoff table

State of nature	Network A	Network B	Network C
State S_1	100	80	60
State S_2	45	50	30
State S_3	40	50	35
State S_4	48	52	40

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Network A}) = \max\{100, 45, 40, 48\} = 100,$$
$$\max(\text{Network B}) = \max\{80, 50, 50, 52\} = 80,$$
$$\max(\text{Network C}) = \max\{60, 30, 35, 40\} = 60.$$

$$\max\{\max(\text{Network A}), \max(\text{Network B}), \max(\text{Network C})\}$$
$$= \max\{100, 80, 60\} = 100.$$

The Maximax choice is Network A because it has the highest possible payoff.

$$\min(\text{Network A}) = \min\{100, 45, 40, 48\} = 40,$$
$$\min(\text{Network B}) = \min\{80, 50, 50, 52\} = 50,$$
$$\min(\text{Network C}) = \min\{60, 30, 35, 40\} = 30.$$

$$\max\{\min(\text{Network A}), \min(\text{Network B}), \min(\text{Network C})\}$$
$$= \max\{40, 50, 30\} = 50.$$

The Maximin choice is Network B because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{State } S_1 : \max\{100, 80, 60\} = 100,$$
$$\text{State } S_2 : \max\{45, 50, 30\} = 50,$$
$$\text{State } S_3 : \max\{40, 50, 35\} = 50,$$
$$\text{State } S_4 : \max\{48, 52, 40\} = 52.$$

Regret table (best payoff − payoff):

Alternative	S_1	S_2	S_3	S_4	Maximum regret
Network A	$100 - 100 = 0$	$50 - 45 = 5$	$50 - 40 = 10$	$52 - 48 = 4$	10
Network B	$100 - 80 = 20$	$50 - 50 = 0$	$50 - 50 = 0$	$52 - 52 = 0$	20
Network C	$100 - 60 = 40$	$50 - 30 = 20$	$50 - 35 = 15$	$52 - 40 = 12$	40

The Minimax Regret choice is Network A because it has the smallest maximum regret, 10.

Decisions summary

Criterion	Final decision
Maximax	Network A
Maximin	Network B
Minimax Regret	Network A

8.3 Inventory Strategy

A retailer must choose among Inventory Strategy A, Strategy B, and Strategy C. Four states of nature represent different demand conditions. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select an inventory strategy.

Payoff table

State of nature	Strategy A	Strategy B	Strategy C
State S_1	100	70	50
State S_2	20	60	40
State S_3	0	50	30
State S_4	25	55	45

Questions/Tasks.

- **C4** Applies the Maximin and Maximax criteria
- **C5** Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

$$\begin{aligned}
\max(\text{Strategy A}) &= \max\{100, 20, 0, 25\} = 100,\\
\max(\text{Strategy B}) &= \max\{70, 60, 50, 55\} = 70,\\
\max(\text{Strategy C}) &= \max\{50, 40, 30, 45\} = 50.
\end{aligned}$$

$$\begin{aligned}
&\max\{\max(\text{Strategy A}), \max(\text{Strategy B}), \max(\text{Strategy C})\}\\
&\quad = \max\{100, 70, 50\} = 100.
\end{aligned}$$

The Maximax choice is Strategy A because it has the highest possible payoff.

$$\begin{aligned}
\min(\text{Strategy A}) &= \min\{100, 20, 0, 25\} = 0,\\
\min(\text{Strategy B}) &= \min\{70, 60, 50, 55\} = 50,\\
\min(\text{Strategy C}) &= \min\{50, 40, 30, 45\} = 30.
\end{aligned}$$

$$\begin{aligned}
&\max\{\min(\text{Strategy A}), \min(\text{Strategy B}), \min(\text{Strategy C})\}\\
&\quad = \max\{0, 50, 30\} = 50.
\end{aligned}$$

The Maximin choice is Strategy B because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\begin{aligned}
\text{State } S_1 &: \max\{100, 70, 50\} = 100,\\
\text{State } S_2 &: \max\{20, 60, 40\} = 60,\\
\text{State } S_3 &: \max\{0, 50, 30\} = 50,\\
\text{State } S_4 &: \max\{25, 55, 45\} = 55.
\end{aligned}$$

Regret table (best payoff $-$ payoff):

Alternative	S_1	S_2	S_3	S_4	Maximum regret
Strategy A	$100 - 100 = 0$	$60 - 20 = 40$	$50 - 0 = 50$	$55 - 25 = 30$	50
Strategy B	$100 - 70 = 30$	$60 - 60 = 0$	$50 - 50 = 0$	$55 - 55 = 0$	30
Strategy C	$100 - 50 = 50$	$60 - 40 = 20$	$50 - 30 = 20$	$55 - 45 = 10$	50

The Minimax Regret choice is Strategy B because it has the smallest maximum regret, 30.

Decisions summary

Criterion	Final decision
Maximax	Strategy A
Maximin	Strategy B
Minimax Regret	Strategy B

8.4 Event Venue

An event company must choose among Venue A, Venue B, and Venue C. Four states of nature represent different attendance levels. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select an event venue.

Payoff table

State of nature	Venue A	Venue B	Venue C
State S_1	100	70	80
State S_2	0	60	50
State S_3	20	55	50
State S_4	40	58	52

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Venue A}) = \max\{100, 0, 20, 40\} = 100,$$
$$\max(\text{Venue B}) = \max\{70, 60, 55, 58\} = 70,$$
$$\max(\text{Venue C}) = \max\{80, 50, 50, 52\} = 80.$$

$$
\begin{aligned}
&\max\{\max(\text{Venue A}), \max(\text{Venue B}), \max(\text{Venue C})\} \\
&= \max\{100, 70, 80\} = 100.
\end{aligned}
$$

The Maximax choice is Venue A because it has the highest possible payoff.

$$
\begin{aligned}
\min(\text{Venue A}) &= \min\{100, 0, 20, 40\} = 0, \\
\min(\text{Venue B}) &= \min\{70, 60, 55, 58\} = 55, \\
\min(\text{Venue C}) &= \min\{80, 50, 50, 52\} = 50.
\end{aligned}
$$

$$
\begin{aligned}
&\max\{\min(\text{Venue A}), \min(\text{Venue B}), \min(\text{Venue C})\} \\
&= \max\{0, 55, 50\} = 55.
\end{aligned}
$$

The Maximin choice is Venue B because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$
\begin{aligned}
\text{State } S_1 &: \max\{100, 70, 80\} = 100, \\
\text{State } S_2 &: \max\{0, 60, 50\} = 60, \\
\text{State } S_3 &: \max\{20, 55, 50\} = 55, \\
\text{State } S_4 &: \max\{40, 58, 52\} = 58.
\end{aligned}
$$

Regret table (best payoff − payoff):

Alternative	S_1	S_2	S_3	S_4	Maximum regret
Venue A	$100 - 100 = 0$	$60 - 0 = 60$	$55 - 20 = 35$	$58 - 40 = 18$	60
Venue B	$100 - 70 = 30$	$60 - 60 = 0$	$55 - 55 = 0$	$58 - 58 = 0$	30
Venue C	$100 - 80 = 20$	$60 - 50 = 10$	$55 - 50 = 5$	$58 - 52 = 6$	20

The Minimax Regret choice is Venue C because it has the smallest maximum regret, 20.

Decisions summary

Criterion	Final decision
Maximax	Venue A
Maximin	Venue B
Minimax Regret	Venue C

8.5 Supply Contract

A company must choose among Supply Contract A, Contract B, and Contract C. Four states of nature represent different market and supply conditions. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a supply contract.

Payoff table

State of nature	Contract A	Contract B	Contract C
State S_1	90	80	70
State S_2	70	50	45
State S_3	60	40	35
State S_4	65	55	50

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

$$\begin{aligned} \max(\text{Contract A}) &= \max\{90, 70, 60, 65\} = 90, \\ \max(\text{Contract B}) &= \max\{80, 50, 40, 55\} = 80, \\ \max(\text{Contract C}) &= \max\{70, 45, 35, 50\} = 70. \end{aligned}$$

$$\begin{aligned} &\max\{\max(\text{Contract A}), \max(\text{Contract B}), \max(\text{Contract C})\} \\ &\quad = \max\{90, 80, 70\} = 90. \end{aligned}$$

The Maximax choice is Contract A because it has the highest possible payoff.

$$\begin{aligned} \min(\text{Contract A}) &= \min\{90, 70, 60, 65\} = 60, \\ \min(\text{Contract B}) &= \min\{80, 50, 40, 55\} = 40, \\ \min(\text{Contract C}) &= \min\{70, 45, 35, 50\} = 35. \end{aligned}$$

$$\begin{aligned} &\max\{\min(\text{Contract A}), \min(\text{Contract B}), \min(\text{Contract C})\} \\ &\quad = \max\{60, 40, 35\} = 60. \end{aligned}$$

The Maximin choice is Contract A because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\begin{aligned} \text{State } S_1 &: \max\{90, 80, 70\} = 90, \\ \text{State } S_2 &: \max\{70, 50, 45\} = 70, \\ \text{State } S_3 &: \max\{60, 40, 35\} = 60, \\ \text{State } S_4 &: \max\{65, 55, 50\} = 65. \end{aligned}$$

Regret table (best payoff $-$ payoff):

Alternative	S_1	S_2	S_3	S_4	Maximum regret
Contract A	$90-90=0$	$70-70=0$	$60-60=0$	$65-65=0$	0
Contract B	$90-80=10$	$70-50=20$	$60-40=20$	$65-55=10$	20
Contract C	$90-70=20$	$70-45=25$	$60-35=25$	$65-50=15$	25

The Minimax Regret choice is Contract A because it has the smallest maximum regret, 0.

Decisions summary

Criterion	Final decision
Maximax	Contract A
Maximin	Contract A
Minimax Regret	Contract A

9 Problems 3×4

Problems:

9.1 Production System

A manufacturing company must choose among Production System A, System B, System C, and System D. Three states of nature represent different operating conditions. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a production system.

Payoff table

State of nature	System A	System B	System C	System D
State S_1	100	70	60	55
State S_2	50	49	30	35
State S_3	50	90	40	45

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

$$\begin{aligned}\max(\text{System A}) &= \max\{100, 50, 50\} = 100,\\ \max(\text{System B}) &= \max\{70, 49, 90\} = 90,\\ \max(\text{System C}) &= \max\{60, 30, 40\} = 60,\\ \max(\text{System D}) &= \max\{55, 35, 45\} = 55.\end{aligned}$$

$$\begin{aligned}&\max\{\max(\text{System A}), \max(\text{System B}), \max(\text{System C}), \max(\text{System D})\}\\ &\quad = \max\{100, 90, 60, 55\} = 100.\end{aligned}$$

The Maximax choice is System A because it has the highest possible payoff.

$$\begin{aligned}\min(\text{System A}) &= \min\{100, 50, 50\} = 50,\\ \min(\text{System B}) &= \min\{70, 49, 90\} = 49,\\ \min(\text{System C}) &= \min\{60, 30, 40\} = 30,\\ \min(\text{System D}) &= \min\{55, 35, 45\} = 35.\end{aligned}$$

$$\begin{aligned}&\max\{\min(\text{System A}), \min(\text{System B}), \min(\text{System C}), \min(\text{System D})\}\\ &\quad = \max\{50, 49, 30, 35\} = 50.\end{aligned}$$

The Maximin choice is System A because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\begin{aligned}\text{State } S_1 &: \max\{100, 70, 60, 55\} = 100,\\ \text{State } S_2 &: \max\{50, 49, 30, 35\} = 50,\\ \text{State } S_3 &: \max\{50, 90, 40, 45\} = 90.\end{aligned}$$

Regret table (best payoff $-$ payoff):

Alternative	S_1	S_2	S_3	Maximum regret
System A	$100 - 100 = 0$	$50 - 50 = 0$	$90 - 50 = 40$	40
System B	$100 - 70 = 30$	$50 - 49 = 1$	$90 - 90 = 0$	30
System C	$100 - 60 = 40$	$50 - 30 = 20$	$90 - 40 = 50$	50
System D	$100 - 55 = 45$	$50 - 35 = 15$	$90 - 45 = 45$	45

The Minimax Regret choice is System B because it has the smallest maximum regret, 30.

Decisions summary

Criterion	Final decision
Maximax	System A
Maximin	System A
Minimax Regret	System B

9.2 Service Network

A service company must choose among Service Network A, Network B, Network C, and Network D. Three states of nature represent different demand conditions. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a service network.

Payoff table

State of nature	Network A	Network B	Network C	Network D
State S_1	100	80	60	70
State S_2	45	50	30	40
State S_3	40	50	35	45

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

$$\begin{aligned}
\max(\text{Network A}) &= \max\{100, 45, 40\} = 100, \\
\max(\text{Network B}) &= \max\{80, 50, 50\} = 80, \\
\max(\text{Network C}) &= \max\{60, 30, 35\} = 60, \\
\max(\text{Network D}) &= \max\{70, 40, 45\} = 70.
\end{aligned}$$

$$\begin{aligned}
&\max\{\max(\text{Network A}), \max(\text{Network B}), \max(\text{Network C}), \max(\text{Network D})\} \\
&\quad = \max\{100, 80, 60, 70\} = 100.
\end{aligned}$$

The Maximax choice is Network A because it has the highest possible payoff.

$$\begin{aligned}
\min(\text{Network A}) &= \min\{100, 45, 40\} = 40, \\
\min(\text{Network B}) &= \min\{80, 50, 50\} = 50, \\
\min(\text{Network C}) &= \min\{60, 30, 35\} = 30, \\
\min(\text{Network D}) &= \min\{70, 40, 45\} = 40.
\end{aligned}$$

$$\begin{aligned}
&\max\{\min(\text{Network A}), \min(\text{Network B}), \min(\text{Network C}), \min(\text{Network D})\} \\
&\quad = \max\{40, 50, 30, 40\} = 50.
\end{aligned}$$

The Maximin choice is Network B because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{State } S_1 : \max\{100, 80, 60, 70\} = 100,$$
$$\text{State } S_2 : \max\{45, 50, 30, 40\} = 50,$$
$$\text{State } S_3 : \max\{40, 50, 35, 45\} = 50.$$

Regret table (best payoff − payoff):

Alternative	S_1	S_2	S_3	Maximum regret
Network A	$100 - 100 = 0$	$50 - 45 = 5$	$50 - 40 = 10$	10
Network B	$100 - 80 = 20$	$50 - 50 = 0$	$50 - 50 = 0$	20
Network C	$100 - 60 = 40$	$50 - 30 = 20$	$50 - 35 = 15$	40
Network D	$100 - 70 = 30$	$50 - 40 = 10$	$50 - 45 = 5$	30

The Minimax Regret choice is Network A because it has the smallest maximum regret, 10.

Decisions summary

Criterion	Final decision
Maximax	Network A
Maximin	Network B
Minimax Regret	Network A

9.3 Inventory Strategy

A retailer must choose among Inventory Strategy A, Strategy B, Strategy C, and Strategy D. Three states of nature represent different demand conditions. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select an inventory strategy.

Payoff table

State of nature	Strategy A	Strategy B	Strategy C	Strategy D
State S_1	100	70	50	60
State S_2	20	60	40	45
State S_3	0	50	30	35

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

$$\begin{aligned}
\max(\text{Strategy A}) &= \max\{100, 20, 0\} = 100, \\
\max(\text{Strategy B}) &= \max\{70, 60, 50\} = 70, \\
\max(\text{Strategy C}) &= \max\{50, 40, 30\} = 50, \\
\max(\text{Strategy D}) &= \max\{60, 45, 35\} = 60.
\end{aligned}$$

$$\begin{aligned}
&\max\{\max(\text{Strategy A}), \max(\text{Strategy B}), \max(\text{Strategy C}), \max(\text{Strategy D})\} \\
&\quad = \max\{100, 70, 50, 60\} = 100.
\end{aligned}$$

The Maximax choice is Strategy A because it has the highest possible payoff.

$$\begin{aligned}
\min(\text{Strategy A}) &= \min\{100, 20, 0\} = 0, \\
\min(\text{Strategy B}) &= \min\{70, 60, 50\} = 50, \\
\min(\text{Strategy C}) &= \min\{50, 40, 30\} = 30, \\
\min(\text{Strategy D}) &= \min\{60, 45, 35\} = 35.
\end{aligned}$$

$$\begin{aligned}
&\max\{\min(\text{Strategy A}), \min(\text{Strategy B}), \min(\text{Strategy C}), \min(\text{Strategy D})\} \\
&\quad = \max\{0, 50, 30, 35\} = 50.
\end{aligned}$$

The Maximin choice is Strategy B because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\begin{aligned}
\text{State } S_1 &: \max\{100, 70, 50, 60\} = 100, \\
\text{State } S_2 &: \max\{20, 60, 40, 45\} = 60, \\
\text{State } S_3 &: \max\{0, 50, 30, 35\} = 50.
\end{aligned}$$

Regret table (best payoff − payoff):

Alternative	S_1	S_2	S_3	Maximum regret
Strategy A	$100 - 100 = 0$	$60 - 20 = 40$	$50 - 0 = 50$	50
Strategy B	$100 - 70 = 30$	$60 - 60 = 0$	$50 - 50 = 0$	30
Strategy C	$100 - 50 = 50$	$60 - 40 = 20$	$50 - 30 = 20$	50
Strategy D	$100 - 60 = 40$	$60 - 45 = 15$	$50 - 35 = 15$	40

The Minimax Regret choice is Strategy B because it has the smallest maximum regret, 30.

Decisions summary

Criterion	Final decision
Maximax	Strategy A
Maximin	Strategy B
Minimax Regret	Strategy B

9.4 Event Venue

An event company must choose among Venue A, Venue B, Venue C, and Venue D. Three states of nature represent different attendance levels. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select an event venue.

Payoff table

State of nature	Venue A	Venue B	Venue C	Venue D
State S_1	100	70	80	60
State S_2	0	60	50	45
State S_3	20	55	50	40

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

$$\begin{aligned}
\max(\text{Venue A}) &= \max\{100, 0, 20\} = 100, \\
\max(\text{Venue B}) &= \max\{70, 60, 55\} = 70, \\
\max(\text{Venue C}) &= \max\{80, 50, 50\} = 80, \\
\max(\text{Venue D}) &= \max\{60, 45, 40\} = 60.
\end{aligned}$$

$$\begin{aligned}
&\max\{\max(\text{Venue A}), \max(\text{Venue B}), \max(\text{Venue C}), \max(\text{Venue D})\} \\
&\qquad = \max\{100, 70, 80, 60\} = 100.
\end{aligned}$$

The Maximax choice is Venue A because it has the highest possible payoff.

$$\begin{aligned}
\min(\text{Venue A}) &= \min\{100, 0, 20\} = 0, \\
\min(\text{Venue B}) &= \min\{70, 60, 55\} = 55, \\
\min(\text{Venue C}) &= \min\{80, 50, 50\} = 50, \\
\min(\text{Venue D}) &= \min\{60, 45, 40\} = 40.
\end{aligned}$$

$$\begin{aligned}
&\max\{\min(\text{Venue A}), \min(\text{Venue B}), \min(\text{Venue C}), \min(\text{Venue D})\} \\
&\qquad = \max\{0, 55, 50, 40\} = 55.
\end{aligned}$$

The Maximin choice is Venue B because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{State } S_1 : \max\{100, 70, 80, 60\} = 100,$$
$$\text{State } S_2 : \max\{0, 60, 50, 45\} = 60,$$
$$\text{State } S_3 : \max\{20, 55, 50, 40\} = 55.$$

Regret table (best payoff − payoff):

Alternative	S_1	S_2	S_3	Maximum regret
Venue A	$100 - 100 = 0$	$60 - 0 = 60$	$55 - 20 = 35$	60
Venue B	$100 - 70 = 30$	$60 - 60 = 0$	$55 - 55 = 0$	30
Venue C	$100 - 80 = 20$	$60 - 50 = 10$	$55 - 50 = 5$	20
Venue D	$100 - 60 = 40$	$60 - 45 = 15$	$55 - 40 = 15$	40

The Minimax Regret choice is Venue C because it has the smallest maximum regret, 20.

Decisions summary

Criterion	Final decision
Maximax	Venue A
Maximin	Venue B
Minimax Regret	Venue C

9.5 Supply Contract

A company must choose among Supply Contract A, Contract B, Contract C, and Contract D. Three states of nature represent different market and supply conditions. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a supply contract.

Payoff table

State of nature	Contract A	Contract B	Contract C	Contract D
State S_1	90	80	70	75
State S_2	70	50	45	55
State S_3	60	40	35	50

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

C4 - Applies the Maximin and Maximax criteria

$$\begin{aligned}
\max(\text{Contract A}) &= \max\{90, 70, 60\} = 90,\\
\max(\text{Contract B}) &= \max\{80, 50, 40\} = 80,\\
\max(\text{Contract C}) &= \max\{70, 45, 35\} = 70,\\
\max(\text{Contract D}) &= \max\{75, 55, 50\} = 75.
\end{aligned}$$

$$\begin{aligned}
&\max\{\max(\text{Contract A}), \max(\text{Contract B}), \max(\text{Contract C}), \max(\text{Contract D})\}\\
&\quad= \max\{90, 80, 70, 75\} = 90.
\end{aligned}$$

The Maximax choice is Contract A because it has the highest possible payoff.

$$\begin{aligned}
\min(\text{Contract A}) &= \min\{90, 70, 60\} = 60,\\
\min(\text{Contract B}) &= \min\{80, 50, 40\} = 40,\\
\min(\text{Contract C}) &= \min\{70, 45, 35\} = 35,\\
\min(\text{Contract D}) &= \min\{75, 55, 50\} = 50.
\end{aligned}$$

$$\begin{aligned}
&\max\{\min(\text{Contract A}), \min(\text{Contract B}), \min(\text{Contract C}), \min(\text{Contract D})\}\\
&\quad= \max\{60, 40, 35, 50\} = 60.
\end{aligned}$$

The Maximin choice is Contract A because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\begin{aligned}
\text{State } S_1 &: \max\{90, 80, 70, 75\} = 90,\\
\text{State } S_2 &: \max\{70, 50, 45, 55\} = 70,\\
\text{State } S_3 &: \max\{60, 40, 35, 50\} = 60.
\end{aligned}$$

Regret table (best payoff − payoff):

Alternative	S_1	S_2	S_3	Maximum regret
Contract A	$90 - 90 = 0$	$70 - 70 = 0$	$60 - 60 = 0$	0
Contract B	$90 - 80 = 10$	$70 - 50 = 20$	$60 - 40 = 20$	20
Contract C	$90 - 70 = 20$	$70 - 45 = 25$	$60 - 35 = 25$	25
Contract D	$90 - 75 = 15$	$70 - 55 = 15$	$60 - 50 = 10$	15

The Minimax Regret choice is Contract A because it has the smallest maximum regret, 0.

Decisions summary

Criterion	Final decision
Maximax	Contract A
Maximin	Contract A
Minimax Regret	Contract A